

Table 3: Semiring families for the temporal and multi-agent layers.

Family	$\oplus/\otimes$	Idemp.	De Morgan.	$\ominus$	Cont.	Interpretation
Boolean	$\vee / \wedge$	✓	✓		✗	qualitative satisfaction
Min-max	$\max / \min$	✓	✓		✓	perturbation margin
Tropical	$\min / +$	$\oplus$ only	✗		✓	cumulative cost
Probabilistic	$+ / \times$	✗	partial		✓	satisfaction likelihood

A Compositional Behavior of Algebraic Semantics

The algebraic semantics of STL-GO are parametric with respect to the underlying algebraic structures used for temporal, graph-based, and multi-agent aggregation. Different choices of these structures induce distinct interpretations of satisfaction, enabling logical, cost-based, and probabilistic reasoning within a unified framework. We summarize the principal design choices below.

A.1 Semiring choices.

The temporal and multi-agent layers are governed by a De Morgan algebra (or semiring) $(K, \oplus, \otimes, \ominus, \perp, \top)$. Different instantiations yield different semantics:

- **Lattice-based semirings.** $(K, \oplus, \otimes) \in \{(\mathbb{B}, \vee, \wedge), (\mathbb{R}^\infty, \max, \min)\}$. Both operations are *idempotent*: $a \oplus a = a$ and $a \otimes a = a$, forming a bounded distributive lattice. A De Morgan negation exists ($\neg$ for Boolean, $-$ for min-max), so the semantics of negation $\rho_i(\sigma, \neg\varphi, t) = \ominus \rho_i(\sigma, \varphi, t)$ are well-defined without restriction. The Boolean algebra yields qualitative satisfaction; the min-max algebra yields the standard STL robustness, where the value measures the perturbation margin [11].
- **Tropical semirings.** $(K, \oplus, \otimes) \in \{(\mathbb{R}^+ \cup \{\infty\}, \min, +), (\mathbb{R} \cup \{-\infty\}, \max, +)\}$. Here $\oplus$ selects the best alternative while $\otimes = +$ accumulates values along a conjunction or temporal sequence, rather than taking a bottleneck. Multiplication is not idempotent ($a + a \neq a$), and no De Morgan negation exists: one cannot define $\ominus$ satisfying $\ominus(a \oplus b) = \ominus a \otimes \ominus b$ when $\oplus$ and $\otimes$ have different algebraic structure.
- **Probabilistic and counting semirings.** $(\mathbb{R}_{\geq 0}, +, \times, 0, 1)$, or its restriction to $[0, 1]$. Neither operation is idempotent: $p + p \neq p$ and $p \times p \neq p$. A negation $\ominus(p) = 1 - p$ can be defined on $[0, 1]$, but it satisfies De Morgan’s laws only under an independence assumption ($\Pr(\neg A) = 1 - \Pr(A)$ is exact, but $\Pr(A \vee B) = \Pr(A) + \Pr(B)$ requires disjointness). The interpretation is probabilistic: conjunction multiplies probabilities and disjunction sums them, giving a satisfaction likelihood under stochastic dynamics. This family is relevant when agent behaviour is modelled stochastically and one seeks to evaluate $\Pr(\sigma \models \phi)$ or an expected robustness.

Table 3 summarises the key properties.

Table 4: Algebraic instantiations of the aggregation pipeline for different semantics. (id = Identity)

Semantics	K	$(\oplus, \otimes)$	M	$\boxplus$	$\mathcal{A}_E$	h
Boolean	$\mathbb{B}$	$(\vee, \wedge)$	$\mathbb{B}$	$\vee$	$\bigvee_{\substack{S \subseteq N_i \\ S =e_1}} \bigwedge_{j \in S} r_j$	id
Min-max	$\mathbb{R}^\infty$	$(\max, \min)$	$\mathbb{R}^\infty$	$\max$	$\min(r_{(e_1)}, -r_{(e_2+1)})$	id
Sign.-def.	$\mathbb{R}^\infty$	$-$	$\mathbb{Z}$	$+$	$\begin{cases} c^+ - e_1, & c^+ < e_1 \\ 0, & e_1 \leq c^+ \leq e_2 \\ e_2 - c^+, & c^+ > e_2 \end{cases}$	id
Hybrid	$\mathbb{R}^\infty$	$(\max, \min)$	$K \times \mathbb{Z}$	$(\max, +)$	(r_{mm}, c^+)	$r + \alpha(c)$

A.2 Graph aggregation choices.

The graph operators aggregate a multiset of neighbour robustness values $\{r_1, \dots, r_m\} \subset K$ into a single value $\mathcal{A}_E^K(r_1, \dots, r_m) \in K$, subject to the count constraint $E = [e_1, e_2]$. This aggregation is *not* fixed by the semiring; we identify some examples, distinguished by the algebraic structure of their underlying aggregation.

- **Idempotent reducers.** [$(K, \max, \min)$ or min-max selection]. Let $r_j = \rho_j(\sigma, \varphi, t)$ denote the robustness of φ at neighbour j , and sort the values in decreasing order: $r_{(1)} \geq r_{(2)} \geq \dots \geq r_{(m)}$, with the convention $r_{(k)} = -\infty$ for $k > m$. For a count interval $E = [e_1, e_2]$, define $\mathcal{A}_{[e_1, e_2]}^{\text{os}}(r_1, \dots, r_m) = \min(r_{(e_1)}, -r_{(e_2+1)})$. The first term $r_{(e_1)}$ is positive iff at least e_1 neighbours have positive robustness; the second term $-r_{(e_2+1)}$ is positive iff at most e_2 do. Their minimum is positive iff both conditions hold, i.e., iff the count of satisfying neighbours lies in $[e_1, e_2]$. The value measures the perturbation margin of the critical neighbour: how much the e_1 -th best neighbour's state can change before the count drops below e_1 . It is sound (i.e. the sign agrees with Boolean verdict), continuous in agent states.
- **Counting-based aggregators.** [$(\mathbb{Z}, +, 0, -)$, an abelian group]. Let $c = |\{j : r_j > 0\}|$ and define $\mathcal{A}_{[e_1, e_2]}^{\text{cd}}(r_1, \dots, r_m) = \min(c - e_1, e_2 - c)$. The value measures the count surplus or deficit: how many neighbours could switch their Boolean verdict before the cardinality constraint breaks. A value of +3 means three. It is *discontinuous* a single neighbour's robustness crossing zero induces a unit jump.
- **Averaging and linear aggregators.** [$(\mathbb{R}, +, \cdot, 0, 1)$ with normalisation by count]. We may define it as: $\mathcal{A}_{[e_1, e_2]}^{\text{avg}}(r_1, \dots, r_m) = \frac{1}{m} \sum_{j=1}^m r_j$. This measures the average quality of neighbour satisfaction, giving a smooth global summary that weights all neighbours equally rather than focusing on the critical one. It does not directly enforce the count constraint E . It is continuous and differentiable, but does not recover the Boolean semantics in general (a positive average does not imply that e_1 neighbours are individually satisfied).

Table 5: Accumulator families for the graph-operator layer.

Family	Sound	Continuous	Semiring -derived	Interpretation
min-max	✓	✓	✓	critical-neighbour margin
Signed-deficit	✓	✗	✗	count surplus/deficit
Averaging	✗	✓	✗	mean neighbour quality
Smooth/fuzzy	approx.	✓	✗	differentiable surrogate

- **Smooth or fuzzy aggregators.** They include parameterised approximations, typically involving exp, log, or sigmoid functions. For e.g., a product-of-sigmoids $\prod_j \sigma(\beta r_j)$ provides a smooth approximation to counting. They are differentiable everywhere, making them amenable to gradient-based optimisation and learning. It has no clean algebraic characterisation; soundness holds only approximately, with the approximation quality controlled by the temperature parameter β .

Table 5 summarises the key properties.

Layer-wise composition. These algebraic choices may be applied uniformly across all layers, or selected independently for temporal, graph, and multi-agent aggregation. A uniform choice yields a consistent interpretation across time, structure, and agents, while heterogeneous choices allow greater flexibility, such as combining logical temporal reasoning with count-based graph aggregation.

B Theoretical Results

B.1 Proof of Theorem 1

Proof. Let $s, s' \in S$ with $s \preceq_S s'$. By definition of the pointwise order on S , this means $s(j) \preceq_K s'(j) \forall j \in \mathcal{V}$.

Fix an agent $i \in \mathcal{V}$ and a graph type $\tau \in \mathcal{T}$. Applying the restriction map to the eligible neighborhood $N_i^{\tau, \circ}(t, W)$ yields $\text{res}_i^\tau(s) \preceq \text{res}_i^\tau(s')$ in the product order on $K^{N_i^{\tau, \circ}(t, W)}$, since restriction simply selects a subset of coordinates and is order-preserving by assumption (1).

By monotonicity of the fold map $\mathcal{A}_E$, we obtain

$$\mathcal{A}_E(\text{res}_i^\tau(s)) \preceq_M \mathcal{A}_E(\text{res}_i^\tau(s')).$$

Applying the monotone readout map $h : M \rightarrow K$ gives

$$h(\mathcal{A}_E(\text{res}_i^\tau(s))) \preceq_K h(\mathcal{A}_E(\text{res}_i^\tau(s'))).$$

Thus, for every graph type τ , the corresponding per-type output is order-preserved. Collecting these values over all graph types produces two tuples in

$K^{|\mathcal{T}|}$, say $(x_\tau)_{\tau \in \mathcal{T}} \preceq (x'_\tau)_{\tau \in \mathcal{T}}$, again in the pointwise order. By monotonicity of the graph-type quantifier $\mathcal{Q}_\#^K$, we conclude that

$$T_i^\#(s) = \mathcal{Q}_\#^K((x_\tau)_{\tau \in \mathcal{T}}) \preceq_K \mathcal{Q}_\#^K((x'_\tau)_{\tau \in \mathcal{T}}) = T_i^\#(s').$$

Hence the local graph operator is monotone.

If the same reasoning applies for every agent $i \in \mathcal{V}$, then $\forall i$,

$$(T(s))(i) = T_i^\#(s) \preceq_K T_i^\#(s') = (T(s'))(i).$$

Therefore $T(s) \preceq_S T(s')$, which proves monotonicity of the global operator $T : S \rightarrow S$. $\square$

B.2 Proof of Proposition 1

Proof. Fix an agent i , time t , graph type τ , and let $\mathbf{r}^\tau = (\rho_j(\sigma, \varphi, t))_{j \in \mathcal{N}_i^{\tau, \circ}(t, W)}$ be the tuple of input values for that graph type.

By Condition (1), $r_j^\tau \succ \perp_K \iff \sigma, j, t \models \varphi$. Hence, $c^+(\mathbf{r}^\tau) = |\{j \in \mathcal{N}_i^{\tau, \circ}(t, W) : \sigma, j, t \models \varphi\}|$, i.e., the number of positive inputs is exactly the number of satisfying neighbors.

Now apply Condition (2): $c^+(\mathbf{r}^\tau) \in [e_1, e_2] \iff F_E(\mathbf{r}^\tau) \succ \perp_K$. Therefore, for each graph type τ , the quantitative output is positive if and only if the Boolean count constraint is satisfied.

For existential graph-type quantification, the operator is $\mathcal{Q}_\exists^K = \bigoplus_{\tau \in \mathcal{T}}$. By sign-correctness of $\bigoplus$, $\bigoplus_{\tau} x_\tau \succ \perp_K \iff \exists \tau : x_\tau \succ \perp_K$. Hence, $\rho_i(\sigma, \mathbf{In}_{\mathcal{G}, E}^{W, \exists} \varphi, t) \succ \perp_K$ if and only if there exists a graph type τ for which the Boolean interval-count condition holds. This is exactly the Boolean semantics of the existential graph-type quantifier.

We can similarly show the case for universal graph-type quantification using the $\bigotimes$ operator. The argument for outgoing operators is identical, replacing $\mathcal{N}_i^{\tau, \text{in}}$ by $\mathcal{N}_i^{\tau, \text{out}}$. Thus the graph operators are sound and complete. $\square$

B.3 Proof for Lemma 1

Proof. By definition of the multi-agent semantics,

$$\rho(\sigma, \mathbf{EX}_\mathcal{V} \varphi, t) = \bigoplus_{i \in \mathcal{V}} \rho_i(\sigma, \varphi, t), \quad \rho(\sigma, \mathbf{FA}_\mathcal{V} \varphi, t) = \bigotimes_{i \in \mathcal{V}} \rho_i(\sigma, \varphi, t).$$

Applying the assumed sign-correctness of finite $\bigoplus$ - and $\bigotimes$ -aggregations yields

$$\begin{aligned} \rho(\sigma, \mathbf{EX}_\mathcal{V} \varphi, t) \succ \perp_K &\iff \exists i \in \mathcal{V} : \rho_i(\sigma, \varphi, t) \succ \perp_K, \\ \rho(\sigma, \mathbf{FA}_\mathcal{V} \varphi, t) \succ \perp_K &\iff \forall i \in \mathcal{V} : \rho_i(\sigma, \varphi, t) \succ \perp_K. \end{aligned}$$

If the local semantics of φ are sound and complete, then

$$\rho_i(\sigma, \varphi, t) \succ \perp_K \Rightarrow \sigma, i, t \models \varphi, \quad \sigma, i, t \models \varphi \Rightarrow \rho_i(\sigma, \varphi, t) \succeq \perp_K.$$

Combining these equivalences with the Boolean semantics of $\mathbf{EX}_\mathcal{V}$ and $\mathbf{FA}_\mathcal{V}$ gives soundness and completeness of the quantified formulas. $\square$

B.4 Proof for Theorem 2

Proof. The proof follows by structural induction on the formula ϕ . *Base case.* If ϕ is an atomic predicate, the result holds by assumption (1).

Inductive step. Assume the claim holds for all immediate subformulas of ϕ .

If ϕ is obtained from its subformulas using a Boolean or temporal operator, then the result follows directly from assumption (2), since the temporal/Boolean fragment is sound and complete over K .

If ϕ is obtained using a graph operator, then by the inductive hypothesis the inputs to the graph aggregation have the correct sign with respect to the Boolean semantics of the subformula. Hence the conditions of Proposition 1 apply, and the graph operator preserves soundness and completeness.

If ϕ is obtained by graph-type quantification or by multi-agent quantification, the result follows from Lemma 1, since these quantifiers preserve threshold semantics. If ϕ is an agent embedding $i.\psi$, then $\rho(\sigma, i.\psi, t) = \rho_i(\sigma, \psi, t)$, and the result follows immediately from the inductive hypothesis.

Since all formation rules preserve soundness and completeness, the result holds for all STL-GO formulas ϕ . $\square$

B.5 General compositional soundness and completeness

Theorem 4. *Let K be a partially ordered set with threshold $\perp_K$, and define the positive and non-positive regions $K^+ := \{x \in K : x \succ \perp_K\}$, $K^- := \{x \in K : x \preceq \perp_K\}$. Consider a compositional semantics in which each formula ϕ is interpreted by a map $\rho(\sigma, \phi, t) \in K$, constructed recursively from: (i) atomic predicate valuations, (ii) a collection of operators $F : K^n \rightarrow K$ applied to subformula values. Assume:*

1. *For every atomic predicate μ , $\rho(\sigma, \mu, t) \in K^+ \iff \sigma, t \models \mu$;*
2. *for every semantic operator $F : K^n \rightarrow K$, there exists a Boolean operator $f : \{0, 1\}^n \rightarrow \{0, 1\}$ such that for all inputs $x_1, \dots, x_n \in K$,*

$$F(x_1, \dots, x_n) \in K^+ \iff f(\mathbf{1}_{x_1 \in K^+}, \dots, \mathbf{1}_{x_n \in K^+}) = 1.$$

Then for every formula ϕ , $\rho(\sigma, \phi, t) \in K^+ \iff \sigma, t \models \phi$. That is, the quantitative semantics are sound and complete with respect to the Boolean semantics.

Proof. The proof is by structural induction on the formula ϕ .

Base case. If ϕ is an atomic predicate, the claim holds by assumption (1).

Inductive step. Suppose $\phi = F(\phi_1, \dots, \phi_n)$ for some operator F . By the induction hypothesis, $\rho(\sigma, \phi_j, t) \in K^+ \iff \sigma, t \models \phi_j$ for all j . Applying assumption (2),

$$\rho(\sigma, \phi, t) = F(\rho(\sigma, \phi_1, t), \dots, \rho(\sigma, \phi_n, t)) \in K^+$$

if and only if $f(\mathbf{1}_{\sigma, t \models \phi_1}, \dots, \mathbf{1}_{\sigma, t \models \phi_n}) = 1$, which is precisely the Boolean semantics of ϕ .

Thus the claim holds for ϕ , completing the induction. $\square$

B.6 Proof of Theorem 3

Proof. Algorithm 1 performs a single bottom-up traversal of the parse tree, visiting each of the $|\varphi|$ subformulae once, and for every subformula computes robustness values for all N agents and T time steps. Predicate, Boolean, temporal, and quantifier nodes require only constant-time algebraic operations per evaluation under the fixed-window assumption (i), yielding $O(NT)$ work per node. For a graph operator, each evaluation at agent i and time t iterates over the M graph types and folds the robustness values of the eligible neighbours $\mathcal{N}_i^{\tau, \circ}(t, W)$, of which there are at most Δ . By assumptions (ii) and (iii), retrieving the neighbour set is free and each accumulator update and readout is $O(1)$, so the fold costs $O(\Delta)$ per graph type and $O(M\Delta)$ per (i, t) pair. The graph-type quantifier $Q_{\#}^K$ then combines the M per-type results in $O(M)$, dominated by the fold. Thus a graph-operator node costs $O(NTM\Delta)$.

Since graph-operator nodes dominate all other node types, summing over the $|\varphi|$ subformulae gives total runtime $O(|\varphi| NTM\Delta)$.⁶ $\square$

C Min-Max Robustness: Instantiation and Limitations

We now instantiate the three-tier algebraic semantics with the min-max algebra $(\mathbb{R}^{\infty}, \max, \min, -, -\infty, +\infty) = (\mathbb{R}^{\infty}, \max, \min, -, -\infty, +\infty)$ and the min-max accumulator, giving concrete robustness formulas for every operator in STL-GO-S. We then examine a family of examples that expose a structural limitation of the min-max accumulator and propose a parameterised alternative.

Ego-centric (STL-GO-S) temporal operators. For agent i , trace σ , and time t :

$$\begin{aligned}
\rho_i(\sigma, \mu, t) &= \nu(\mathcal{S}_t, i, \mu), \\
\rho_i(\sigma, \neg\varphi, t) &= -\rho_i(\sigma, \varphi, t), \\
\rho_i(\sigma, \varphi_1 \wedge \varphi_2, t) &= \min(\rho_i(\sigma, \varphi_1, t), \rho_i(\sigma, \varphi_2, t)), \\
\rho_i(\sigma, \varphi_1 \vee \varphi_2, t) &= \max(\rho_i(\sigma, \varphi_1, t), \rho_i(\sigma, \varphi_2, t)), \\
\rho_i(\sigma, \mathbf{F}_{[t_1, t_2]} \varphi, t) &= \max_{t' \in t+[t_1, t_2]} \rho_i(\sigma, \varphi, t'), \\
\rho_i(\sigma, \mathbf{G}_{[t_1, t_2]} \varphi, t) &= \min_{t' \in t+[t_1, t_2]} \rho_i(\sigma, \varphi, t'), \\
\rho_i(\sigma, \varphi_1 \mathbf{U}_{[t_1, t_2]} \varphi_2, t) &= \max_{t' \in t+[t_1, t_2]} \min \left(\rho_i(\sigma, \varphi_2, t'), \min_{t'' \in [t, t']} \rho_i(\sigma, \varphi_1, t'') \right). \quad (4)
\end{aligned}$$

The labelling function $\nu(\mathcal{S}_t, i, \mu)$ returns the *signed distance* of agent i 's state from the predicate boundary: positive if $\mu(\mathbf{x}_t^i)$ holds, negative otherwise, with magnitude proportional to the margin.

Graph operators. Fix agent i , time t , graph type τ , and let $N = \mathcal{N}_i^{\tau, \text{in}}(t, W)$ be the set of qualifying incoming neighbours (those connected to i by an edge

⁶ Since $\Delta \leq N - 1$ always, this gives the worst-case bound $O(|\varphi| N^2 T M)$; for sparse interaction graphs $\Delta \ll N$ and the cost is substantially lower.

Table 6: Neighbour robustness configurations ($k=5$, $|N|=6$).

Neighbour robustness values	
S_1 {+10, +10, +10, +10, +10, +10}	S_1 – S_3 satisfy the constraint ($c^+ \geq 5$);
S_2 {+10, +10, +10, +10, +10, -10}	S_4 – S_6 violate it ($c^+ < 5$). Ideally, the
S_3 {+10, +10, +10, +10, +0.5, -10}	robustness should be:
S_4 {+10, +10, +10, +10, -10, -10}	– positive for S_1 – S_3 , with $S_1 >$
S_5 {+10, +10, +10, -10, -10, -10}	$S_2 > S_3 > 0$;
S_6 {+10, +10, -10, -10, -10, -10}	– negative and decreasingly so from
	S_4 through S_6 , reflecting the grow-
	ing count deficit.

with weight in W). Let $r_j = \rho_j(\sigma, \varphi, t)$ for each $j \in N$, and sort these values as $r_{(1)} \geq \dots \geq r_{(|N|)}$, with the convention $r_{(k)} = -\infty$ for $k > |N|$. Then:

$$\rho_i(\sigma, \mathbf{In}_{\mathcal{G}, E}^{W, \exists} \varphi, t) = \max_{\tau: \mathcal{G}^\tau \in \mathbf{G}_t} \min(r_{(e_1)}^\tau, -r_{(e_2+1)}^\tau), \quad (5)$$

$$\rho_i(\sigma, \mathbf{In}_{\mathcal{G}, E}^{W, \forall} \varphi, t) = \min_{\tau: \mathcal{G}^\tau \in \mathbf{G}_t} \min(r_{(e_1)}^\tau, -r_{(e_2+1)}^\tau). \quad (6)$$

The outgoing operators $\mathbf{Out}_{\mathcal{G}, E}^{W, \#}$ are identical with $\mathcal{N}_i^{\tau, \text{out}}$ in place of $\mathcal{N}_i^{\tau, \text{in}}$.
Multi-agent operators.

$$\rho(\sigma, \mathbf{FA}_V \varphi, t) = \min_{i \in V} \rho_i(\sigma, \varphi, t), \quad \rho(\sigma, \mathbf{EX}_V \varphi, t) = \max_{i \in V} \rho_i(\sigma, \varphi, t). \quad (7)$$

The remaining multi-agent temporal operators ($\neg$, $\wedge$, $\vee$, $\mathbf{U}_{[t_1, t_2]}$) follow the same min/max structure as the STL-GO-S operators.

Limitations

The min-max accumulator reports a single value: the margin of the k -th best neighbour, regardless of *how many* neighbours are satisfying or violating. This makes it insensitive to the *count deficit*: configurations that differ dramatically in the number of violating neighbours can receive identical robustness.

Example. Fix $|N| = 6$ neighbours and the threshold $k = e_1 = 5$ (“at least 5 neighbours satisfy ψ ”). Consider the neighbour robustness multisets in Table 6: S_1 has all six neighbours satisfying robustly; S_2 has exactly five at margin +10; S_3 also has five satisfying, but the 5th-best has margin only +0.5 (a much more fragile satisfaction); S_4 – S_6 have 4, 3, 2 satisfying neighbours respectively, each at ± 10 .

Under the (i) **min-max accumulator**, S_1 and S_2 both receive +10: the accumulator cannot distinguish having one spare neighbour from having zero. S_3 receives +0.5, correctly reflecting its fragile 5th neighbour, but S_4 through S_6 all receive -10: a configuration missing one neighbour is indistinguishable from one missing five. The order statistic captures *how robustly* the critical neighbour satisfies ψ , but is entirely blind to *how many* neighbours are satisfying or missing. Under the (ii) **signed-deficit accumulator** ($c^+ - k$), the values +1, 0, 0, -1, -2, -3, -4, -5 provide a monotone gradation reflecting the count surplus or deficit.

Table 7: Accumulator comparison ($k=5$, $|N|=6$). $\checkmark$: satisfies; $\times$: violates.

Set	Bool.	c^+	Min-max.	Sgn-def.	Hybrid	$r_{(k)} + \alpha(c^+ - k)$		
			$r_{(5)}$	$c^+ - k$	$\alpha=1$	$\alpha=5$	$\alpha=10$	
S_1	✓	6	+10	+1	+11	+15	+20	
S_2	✓	5	+10	0	+10	+10	+10	
S_3	✓	5	+0.5	0	+0.5	+0.5	+0.5	
S_4	×	4	-10	-1	-11	-15	-20	
S_5	×	3	-10	-2	-12	-20	-30	
S_6	×	2	-10	-3	-13	-25	-40	

However, S_2 and S_3 both receive exactly 0 despite S_2 being far more robust than S_3 —the signed deficit discards all signal-margin information. We propose a (iii) **hybrid accumulator**: $r_{(k)} + \alpha(c^+ - k)$, combining the order statistic for signal-level margin with the scaled count deficit for sensitivity to the number of satisfying neighbours. The parameter $\alpha > 0$ controls the *exchange rate* between signal margin (in robustness units) and count margin (in number of agents). As shown in Table 7: (a) $S_1 > S_2 > S_3 > 0$ for all $\alpha > 0$: the surplus neighbour in S_1 is reflected as a bonus of $+\alpha$, and S_2 vs. S_3 are distinguished by the critical neighbour’s margin (+10 vs. +0.5). (b) S_4 through S_6 are strictly decreasing, with the spacing growing with α : larger α amplifies the count-deficit penalty. (c) Within a fixed deficit level, the min-max term $r_{(k)}$ still distinguishes configurations by the margin of the critical neighbour.